

LAKENHEATH, United Kingdom

WMO: 035830

Lat: 52.417N

Lon: 0.567E

Elev: 10

StdP: 101.20

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 35034

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-4.1	-2.9	-6.0	2.3	-1.7	-4.8	2.5	-1.0	13.9	10.2	12.4	10.1	1.7	230	0.607

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.9	27.9	18.8	25.7	17.9	23.9	17.2	20.2	25.1	19.1	23.6	18.2	22.1	4.5	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.8	13.7	21.4	17.7	12.7	20.3	16.8	12.0	19.6	57.9	24.8	54.3	23.4	51.4	21.9	25.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.5	10.2	9.1	DB	-7.3	31.5	2.2	2.2	-8.9	33.1	-10.2	34.3	-11.5	35.5	-13.1	37.1	(4)
(5)			WB	-7.7	22.0	2.3	1.5	-9.4	23.1	-10.7	24.0	-12.0	24.8	-13.7	25.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	10.6	4.4	5.0	6.8	9.3	12.5	15.3	17.9	17.4	14.6	11.4	7.4	5.2
	DBStd	5.69	3.42	3.49	3.00	3.35	3.18	2.78	2.65	2.69	2.66	3.22	3.46	3.89
	HDD10.0	754	174	142	105	53	12	1	0	0	1	22	91	154
	HDD18.3	2871	431	374	357	272	182	98	41	50	117	214	327	408
	CDD10.0	988	2	2	7	31	89	159	245	231	137	66	13	5
	CDD18.3	62	0	0	0	0	1	7	28	22	4	1	0	0
	CDH23.3	513	0	0	0	3	17	68	235	160	25	5	0	0
	CDH26.7	114	0	0	0	0	1	13	56	41	4	1	0	0
Wind	WSAvg	4.3	4.8	4.9	4.8	4.2	4.3	4.1	3.9	3.9	3.8	4.0	4.3	4.7
Precipitation	PrecAvg	628	52	42	39	41	48	55	58	65	51	62	63	52
	PrecMax	793	95	78	89	123	117	158	115	151	113	115	109	99
	PrecMin	416	9	7	4	0	12	14	21	7	11	11	20	20
	PrecStd	107	23	19	21	30	27	36	26	36	27	26	23	23
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.7	15.1	17.2	23.1	25.2	28.8	31.3	30.3	26.6	22.2	16.2	14.4
		MCWB	11.9	11.7	12.1	14.6	16.6	19.4	20.6	19.9	18.6	17.2	14.6	12.2
	2%	DB	12.2	12.2	15.2	19.4	23.0	25.1	27.9	27.5	23.0	19.1	15.0	13.2
		MCWB	10.8	9.9	11.7	13.4	15.9	17.9	18.7	18.9	17.5	16.1	13.2	11.8
	5%	DB	11.0	11.2	13.8	17.1	20.9	23.0	25.9	24.8	21.0	17.5	13.9	12.1
		MCWB	9.7	9.4	10.7	12.2	15.1	16.6	18.0	17.9	16.4	15.1	12.2	10.9
	10%	DB	9.9	10.1	12.0	15.2	18.8	21.1	23.9	22.9	19.6	16.1	12.8	11.0
		MCWB	8.7	8.6	9.5	11.2	13.9	15.9	17.5	16.8	15.6	14.2	11.3	10.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.5	12.0	13.8	15.7	18.8	20.9	21.8	21.1	20.6	18.4	15.2	13.1
		MCDB	13.1	13.3	15.8	21.3	22.4	26.4	26.8	27.2	23.9	20.7	16.0	13.6
	2%	WB	11.3	10.7	12.3	14.1	17.1	19.0	20.2	19.7	18.7	16.8	13.7	12.1
		MCDB	12.3	12.0	14.4	17.9	20.6	22.7	25.4	24.8	21.6	18.6	14.6	13.1
	5%	WB	10.0	9.8	11.1	13.0	15.8	17.6	19.1	18.7	17.5	15.7	12.5	11.2
		MCDB	10.9	11.0	13.1	16.3	19.7	21.2	24.1	23.3	20.1	17.2	13.5	12.0
	10%	WB	8.8	8.8	10.0	11.7	14.5	16.5	18.0	17.8	16.4	14.7	11.4	10.1
		MCDB	9.6	10.0	11.8	14.8	18.1	20.3	22.4	21.6	18.5	16.0	12.4	11.0
Mean Daily Temperature Range	5% DB	MDBR	5.6	6.6	7.9	9.5	9.6	9.3	9.9	9.5	9.0	7.8	6.3	5.5
		MCDBR	6.3	7.4	10.1	12.2	13.1	12.7	13.3	12.2	10.8	8.2	6.0	5.7
		MCWBR	5.9	5.8	6.7	7.2	7.0	6.5	5.9	5.4	6.2	5.5	5.0	5.3
	5% WB	MCDBR	6.4	6.9	9.0	10.6	11.4	10.8	11.4	10.7	9.3	7.2	6.2	5.8
		MCWBR	6.3	5.9	6.7	6.8	6.7	6.4	6.0	5.3	6.2	5.5	5.6	5.6
Clear-Sky Solar Irradiance	taub	0.327	0.344	0.379	0.403	0.407	0.411	0.410	0.399	0.377	0.358	0.331	0.317	
	taud	2.354	2.369	2.264	2.212	2.246	2.256	2.271	2.323	2.381	2.417	2.401	2.341	
	Ebn at Noon	659	763	802	828	844	842	835	825	799	737	651	599	
	Edh at Noon	61	80	107	127	129	129	125	113	95	76	58	53	
All-Sky Solar Radiation	RadAvg	0.73	1.41	2.52	3.97	4.80	5.11	4.88	4.11	3.09	1.80	0.91	0.58	
	RadStd	0.07	0.21	0.28	0.41	0.36	0.40	0.51	0.29	0.25	0.16	0.10	0.07	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (77 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page